

Structural functionalization of industrial softwood kraft lignin for simple dip-coating of urea as highly efficient nitrogen fertilizer

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ABSTRACT

Urea coating was conducted using polylactic acid (PLA) blended with industrial softwood kraft lignin after applying a dip-coating technique. The lignin was pre-functionalized via esterification that increased coat layer hydrophobicity or via amination that created an organically bound nitrogen structure. The PLA film reference had many pores, while the film from the PLA-lignin derivative complex was highly homogeneous and had no pores. The coat thickness was generally adjustable by repeating the coating process reaching up to 81% weight against the urea core. After coating, urea release in water was largely delayed, 20–30 and 6–10 times as long as that of uncoated urea or PLA-coated urea respectively. The coated urea will be a highly effective nitrogen fertilizer due to the controlled release after coating, the slow release from the organically bound nitrogen structure, and the expected extra stability against urease hydrolysis and microorganism nitrification after using the lignin structure.

1. Introduction

Nitrogen (N) is an essential element for plant growth and reproduction. Currently, many N fertilizers are utilized for the cultivation of crops and forests, and urea is the most widely used form. However, due to its high water solubility, urea is immediately released after application and substantially leaches into ground water. Moreover, urea cannot be directly absorbed by plants. Conversion to plant-available N is necessary before the N fertilizer can be taken up by the plants, either as NH_4^+ or as NO_3^- after urease hydrolysis or soil microorganism nitrification. However, both conversions are accompanied by further N loss via, for example, ammonia gas evolution from NH_4^+ or water leaching of NO_3^- . Therefore, a large quantity of fertilizer N cannot be eventually utilized by crops and forests but instead escapes to the environment, causing large economic losses and serious pollution. There is thus a demand for the development of more efficient N fertilizers (Wu et al., 2008)(see (Mengel et al., 2001; Pereira et al., 2015)).

In principle, the N utilization efficiency could be improved by decreasing the N release and/or suppression of its conversion to NH_4^+ and NO_3^- in order to supply N in the manner that is best suited for plant absorption and periodic physiological growth. Multiple countermeasures have been considered (see (Chien et al., 2009)). In the

literature, the most commonly applied physical approach has been to process the urea via coating (Chien et al., 2009) or film or membrane encapsulation (Guo et al., 2005; Niu and Li, 2012; Zhang et al., 2014). The coating shell supplies a physical barrier around the urea core against contact with water so that the urea dissolution and release are delayed, and this coated urea is termed controlled release fertilizer (see Ref. (Chien et al., 2009)). Numerous types of synthetic polymers (including instant polymerization (Liang and Liu, 2006)) have been examined as the coating material, but biodegradable or natural polymers are more attractive following the global development trend for environmental preservation. Polylactic acid (PLA) has been reported to be the best polymer among many polymers tested (Devassine et al., 2002). Currently, there are many different coating methods (Kenneth and Allan, 1994a) and techniques (Kenneth and Allan, 1994b) available, but they are usually sophisticated and require special and expensive mechanical equipment, such as fluidized beds and pan coaters (Shipway, 2006). Coating urea with PLA was achieved using the so-called atomizing method, where expensive equipment was used (Papangkorn et al., 2008). A second chemical approach that was reported involves replacing the inorganic urea fertilizer with organically bound N structures, which slows the time for soil conversion into plant-available N, and this organic N structure is termed slow release fertilizer (see (Chien

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et al., 2009)). A third biological approach has been applied to suppress the biological conversion of N-fertilizer via inclusion of chemical additives that possess inhibitory activities into the fertilizer composition, and this compound fertilizer is termed stabilized fertilizer (Chien et al., 2009). Multifunctional slow-release fertilizers have also been subjects of research in the literature (Ni et al., 2009; Wang et al., 2014b; Wang et al., 2012).

Lignin is the second most abundant bio-renewable and biodegradable polymer in nature. It has therefore been chosen for investigation in modified N fertilizer using different approaches. Following the physical approach, lignin has been used as a mono-component in urea coatings. The coating was administered to prepare a pellet of controlled release fertilizer either via simple co-melting of the lignin and the urea powders, followed by pelletization (Basile, 1997), or via directly coating the lignin onto the surface of the urea pellets using coating equipment (Mulder et al., 2011; Wang et al., 2003). However the coating effects were limited. To improve the effects, lignin has been examined as a second layer over an initial polymer layer including PLA (Kannan et al., 2015). In addition, lignin has also been used for complex coating together with, for example, ethylcellulose (Fernández-pérez et al., 2008), rosin (García et al., 1997), starch (Majeed et al., 2016) or wax (Cui et al., 2013). However, all these applications were in combinations of the specific equipment requirements. Following the chemical approach, lignin was processed into organically bound N fertilizer, i.e., into so-called slow release fertilizer, via lignin ammoxidation (Meier et al., 1994; Radlein, 1995), where the N fertilizer is in the form of different amides, such as formamide, acetamide, benzamide, etc., as well as other organic N structures (Potthast et al., 1996). Following the biological approach, lignin itself is generally known for having inhibitory effects against both urease hydrolysis and soil nitrification (Huang et al., 2003; Peng et al., 2004). These functions are believed to originate from its multi-phenolic or polyphenolic functional groups (Lodhi and Killingbeck, 1980), which are present in large amounts in lignin structures. For example, urease suppression has been reported from a lignin study (Wang et al., 2003). In fact, the ammoxidated lignin also showed function in urease hydrolysis inhibition (Chen et al., 2009).

The aim of this study is to develop a highly efficient N fertilizer. The work was based on the urea coating approach using PLA as a basic coating material. In combination with PLA, an industrial softwood kraft lignin has been used for complex coating to further improve the coating effect, and for the first time, coating was conducted without using expensive coating equipment but instead was accomplished using a simple and cheap dip-coating technique. Lignin was utilized after structural functionalization by either esterification or the Mannich reaction. The former rendered a lignin derivative with higher hydrophobicity, imparting improved water barrier properties of the coating and thus higher controlled release efficiency. The latter introduced organically bound N onto the lignin derivative structure, becoming itself a slow release N fertilizer. It could also be expected that adding lignin derivatives to the coating composition would impart lignin's inherent inhibitory effects against both urease hydrolysis and soil nitrification to the coated urea fertilizer. Therefore, by conducting the dip-coating of urea using a PLA-functionalized lignin complex, we demonstrated a simple and cheap coating technique for attaining highly efficient N fertilizer via a single coating that simultaneously supplies physical, chemical and biological effects.

2. Materials and methods

2.1. Materials

The softwood kraft lignin (SKL) was prepared on the pilot scale from the black liquor collected from industrial kraft cooking of softwood, following the LignoBoost process (Oehman et al., 2006). It was directly utilized without any further refinement. Comprehensive structural characterizations of SKL have been conducted and reported earlier

(Wang et al., 2017). Commercial urea pellets and palmitic anhydride (90–95%) were purchased from Sigma-Aldrich. Polylactic acid (PLA) fiber (PL80) with a declared molecular mass distribution (M_w) of 140567 g/mol was obtained from Unitika Ltd. (Osaka, Japan).

2.2. Derivatization

Esterification was accomplished using Ac_2O or palmitic anhydride together with pyridine as the solvent and catalyst. Following the procedure reported by Gellerstedt (1992), the acetylation of SKL into acetylated SKL (AcSKL) was conducted using 500 mg SKL and 5 ml Ac_2O /pyridine (1:1, v/v) at room temperature for 24 h. Excess anhydride reagent was then decomposed by the addition of water and methanol, followed by evaporation under reduced pressure using first toluene to assist in the removal of the acetic acid that was formed and then methanol to remove residual toluene. The palmitoylation of SKL into palmitoylated SKL (PaSKL) was conducted using 200 mg SKL (~ 1 mmol by using 200 g/mol for one lignin aromatic unit), 2 g palmitic anhydride (~ 4 mmol) and 5 ml pyridine at room temperature for 24 h, followed by reflux for 1 h. Excess anhydride reagent was then decomposed by the addition of water and methanol, followed by filtration and thorough washing with water to a neutral pH, and finally freeze-drying.

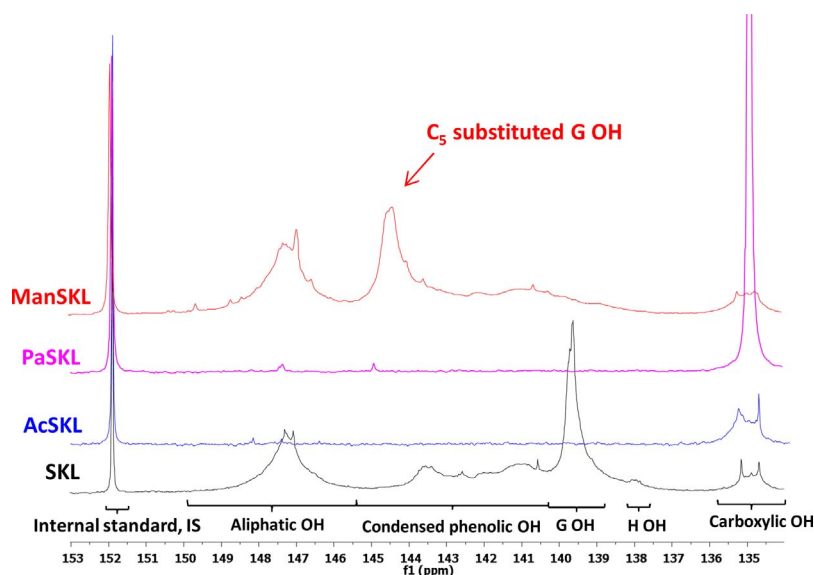
The Mannich reaction of SKL into the product termed ManSKL was conducted according to previous reports (Wang et al., 2017). Briefly, 100 mg SKL (~ 0.50 mmol) was dissolved in 5.0 ml dioxane, followed by the addition of 5.5 mmol diethylamine and 5.5 mmol formaldehyde. The mixture was sealed and stirred at 60 °C for 4 h. After the reaction was complete, the solution of products and unconsumed reagents was evaporated under reduced pressure.

2.3. Chemical analysis

The OH groups of lignin were phosphorylated and quantified by ^{31}P NMR spectroscopy, following a procedure adopted from the literature (Granata and Argyropoulos, 1995). In detail, a dried lignin sample or its functionalized product (30 mg) was dissolved in a mixture composed of 100 μl pyridine/N,N-dimethylformamide (1:1), 100 μl N-hydroxy-5-norbornene-2,3-dicarboxylic acid imide pyridine solution (20 mg ml^{-1}) as an internal standard (IS) and 100 μl chromium acetylacetonate pyridine solution (5 mg ml^{-1}) as a relaxation agent. The reagent 2-chloro-4,4,5,5-tetramethyl-1,3,2-dioxaphospholane (100 μl) was then added, the mixture was stirred for 10 min for completion of the phosphorylation reaction, and then 400 μl chloroform- d was added. ^{31}P NMR spectra were recorded on a Bruker DRX-400 spectrometer (Bruker, Bremen, Germany), and the data were analyzed by the Topspin 3.0 software (Bruker, Bremen, Germany). All chemical shifts are reported relative to the signal of the IS (151.9 ppm). The content of hydroxyl groups was obtained by integration of the following spectral regions: aliphatic OH (150.0–145.4 ppm), condensed phenolic OH, including syringyl OH, (145.4–140.2 ppm), guaiacyl phenolic OH (140.2–139.0 ppm), *p*-hydroxyphenyl phenolic OH (136.5–138.6 ppm), and carboxylic OH (136.0–133.6 ppm).

FTIR analysis was conducted using a Perkin-Elmer Spectrum 2000 FTIR spectrometer (Waltham, MA, USA) that was equipped with an ATR system and a Specac MKII Golden Gate (Creechstone Ridge, GA, USA), scanning from 600 to 4000 cm^{-1} in 1 cm^{-1} intervals with 4 cm^{-1} resolution over 16 scans at room temperature. Before FTIR data collection, a background scan was performed for background correction.

Quantitative carbon (^{13}C NMR) and hydrogen-carbon 2D heteronuclear single quantum coherence (^1H – ^{13}C 2DHSQC NMR) nuclear magnetic resonance analyses were performed on a Bruker DRX-400 spectrometer (Bruker, Bremen, Germany) after dissolving the sample in dimethyl sulfoxide- d_6 or chloroform- d (for palmitoylated SKL) using standard Bruker pulse programs. The data were analyzed by the

Fig. 1. ^{31}P NMR spectra of original SKL, AcSKL, PaSKL and ManSKL.

Topspin 3.0 software (Bruker, Bremen, Germany).

The nitrogen content was determined by dissolving the Mannich reaction product in 0.1 M NaOH with 4% acetone, followed by analysis on a Shimadzu TNM-1 Total Nitrogen Module (Shimadzu, Tokyo, Japan). The system was calibrated with 1–100 ppm KNO_3 standard solutions.

2.4. Film casting

PLA was dissolved in dichloromethane at a concentration of 60% (w/v), and lignin or lignin derivative was dissolved in dioxane at a concentration of 6% (w/v). The PLA-lignin complex was made by mixing both solutions in a volume ratio of 1:1. The films were prepared directly by casting the mixture on Teflon Petri dishes and drying at room temperature overnight. The thickness of the films was 50–70 μm .

2.5. Coating

Dip-coating was performed by immersing ~ 1 g urea pellets with diameters of 1–2 mm, obtained after screening the commercial pellets using sieves, in ~ 10 ml of the above-mentioned 1:1 mixed solution of PLA and lignin for five minutes. The urea pellets were removed from solution and placed on a Teflon film surface after careful separation of the pellets one from another. They were dried in air for > 1 h. The dip-coating was repeated a different number of times to obtain different coating thicknesses.

2.6. Urea release determination

The coated urea was enclosed in a nylon membrane (pore size around 0.1–0.2 mm) folder and placed in a beaker with deionized water that immersed the sample completely. Forced circulation of the water was accomplished using a pump at a flow rate of 30 ml/min through a 1 cm UV determination cell for continuous UV monitoring at 200 nm against time using a UV–vis spectrophotometer (Shimadzu UV2550). This 200 nm absorption method to quantify urea was adapted from (Filutowicz et al., 2004). The maximum reading was then set as 100% release.

2.7. Characterization

2.7.1. Contact angle

The pendant drop method was used and was conducted at 23 $^\circ\text{C}$

with a relative humidity (RH) of 50% using a SL100 B contact angle meter from Solon Information Technology Co., Ltd. (Shanghai, China). The contact angle was recorded at 60 s when the reading was stable.

2.7.2. Scanning electron microscopy (SEM)

SEM analyses were performed using either a Hitachi S-4800 field emission scanning electron microscope (SEM) or a Hitachi TM-1000 Tabletop SEM. For the former, the samples were coated with gold using a Cressington 208HR high-resolution sputter coater before analysis, and the coating thickness (3–5 nm) was controlled by a Cressington thickness monitor controller. For the latter, the samples were directly mounted on the SEM plate for analysis. A pore size analysis was performed using the public domain ImageJ software developed by the US National Institutes of Health (NIH) (available at <http://rsb.info.nih.gov/ij/>) to measure the pores selected on the SEM image from each type of sample.

3. Results and discussion

3.1. Lignin structure

To keep the advantage of low cost, industrial softwood kraft lignin (SKL) was used directly without any further refinement. Nevertheless, as analyzed earlier and reported previously (Wang et al., 2017), this SKL is of high purity, with 99.7% lignin and a low content of ash (0.35%) and carbohydrates (1.2%). It is a typical macromolecular lignin with a molecular mass distribution (MMD) value of 5500 g/mol (M_w) and a polydispersity index (PDI) of 3.9. Compared to the high molecular mass ($M_w = 140567$ g/mol) of the PLA that was used, this lignin could be expected to function as a plasticizer in the PLA-lignin complex coating.

Chemically there are many functional groups present in SKL that can be used for structural derivatization and functionalization. As learned from ^{31}P NMR analysis (Fig. 1 and Table 1, see also the experimental section), the lignin contains aliphatic hydroxyl groups (1.91 mmol/g) on its side chains (at the C_α and C_γ positions, see Fig. 2) and phenolic hydroxyls on its different aromatic rings, mainly from the guaiacyl (G) (2.32 mmol/g) units (at the C_4 position, see Fig. 2), along with small amounts from *p*-hydroxyphenyl (H) (0.24 mmol/g) and different condensed phenolic (2.01 mmol/g) (including a small amount of syringyl (S)) units. There is also a certain amount of carboxylic groups (0.46 mmol/g) present in the lignin. In total, all the aliphatic OHs and up to 4.6 mmol/g of the different aromatic OHs are susceptible

Table 1

Contents (mmol/g) of aliphatic, aromatic (guaiacyl (G), *p*-hydroxyphenyl (H), and condensed phenolic (including a small amount of syringyl)), and carboxyl hydroxyl groups of softwood kraft lignin (SKL) before and after esterification (AcSKL or PaSKL) or the Mannich reaction (ManSKL), as determined by ^{31}P NMR.

Sample	Aliphatic OH	Aromatic OH			Carboxyl OH
		G-OH	H-OH	Condensed phenol-OH	
SKL ^a	1.91 ± 0.11	2.32 ± 0.03	0.24 ± 0.05	2.01 ± 0.04	0.46 ± 0.01
AcSKL	~0	~0	~0	~0	0.43
PaSKL	~0	~0	~0	~0	2.86
ManSKL	1.66	~0	~0	2.97	0.30

^a duplicated determination.

to esterification by different fatty acids (Fig. 2). In addition, the dominant phenolic G units offer a certain amount of reactive sites at the C₅ position of the phenolic aromatic units, amounting to 2.32 mmol/g, which are available for the Mannich reaction (Fig. 2, see (Wang et al., 2017)). Based on the analytical results from the earlier (Wang et al., 2017) and present chemical characterizations, esterification and amination by Mannich reaction are conclusively considered as applicable for structural functionalizations of the SKL.

3.2. Structural functionalization

The first type of functionalization conducted is esterification using acetic anhydride or palmitic anhydride. After esterification, fatty acids were coupled to SKL with a short aliphatic chain ($\text{R} = \text{CH}_3$) in the case of acetic acid or a long chain ($\text{R} = \text{CH}_3(\text{CH}_2)_{14}$) in the case of palmitic acid (Fig. 2). This will increase the hydrophobicity of the lignin derivatives, especially for long-chain palmitic acid, which can be expected to improve the water barrier properties of the PLA-lignin complex coating. Experimentally, acetylation was conducted using acetyl anhydride together with pyridine as both the solvent and catalyst, following the well-known procedure for complete acetylation reaction (Gellerstedt, 1992). Palmitoylation was performed similarly by

replacing the acetyl anhydride with palmitic anhydride.

The obtained lignin derivatives were then comprehensively analyzed using different spectroscopic methods. As revealed by ^{31}P NMR analysis (Fig. 1 and Table 1), all aliphatic and aromatic hydroxyls were completely esterified (corresponding OH peaks disappeared) upon acetylation, while the carboxyl hydroxyls were intact (with a similar quantity as the starting SKL after evaporative removal of all acetic acid formed during the decomposition of excess acetyl anhydride). Correspondingly, as shown in the FTIR spectra (Fig. 3), the signals at approximately 3300 cm^{-1} , generated from the various original OHs, vanished almost completely after acetylation. Two ester carboxyl peaks emerged at 1765 and 1740 cm^{-1} . As described previously (Glasser and Jain, 1993), they are from aromatic (C₄ position) and aliphatic (C_α and C_γ positions, see Fig. 2) esters, respectively. These two types of esters are also shown in the ^{13}C NMR spectra at approximately 170 ppm (Fig. 4) (Faix et al., 1994). Since the esterification occurred at both the C_α and C_γ positions and C₄ position on the aromatic rings, the corresponding signal shifts could be observed in the 2D NMR spectra for C–H e.g. directly at C_α and C_γ positions and at the C₅ position neighboring to C₄ (Fig. 5).

The palmitoylation was complete as indicated by the disappeared OH peaks (Fig. 1 and Table 1). After the reaction, the carboxyl hydroxyl

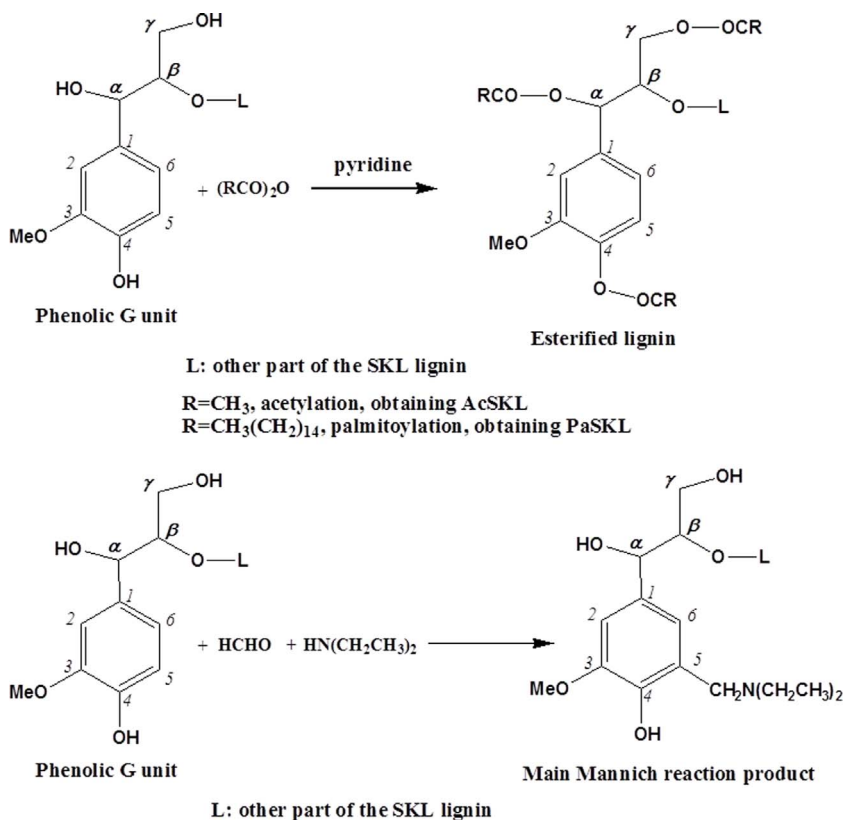


Fig. 2. Lignin functionalization by esterification (top) or the Mannich reaction (bottom), obtaining AcSKL and PaSKL or ManSKL.

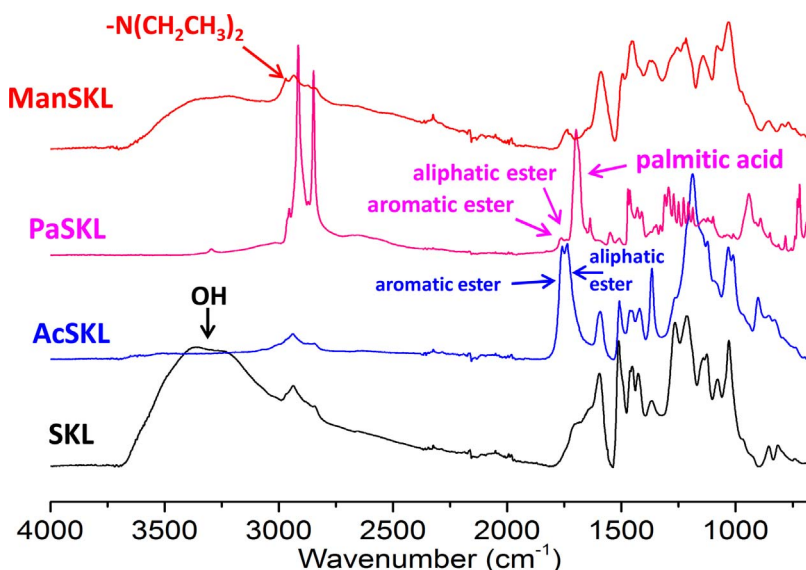


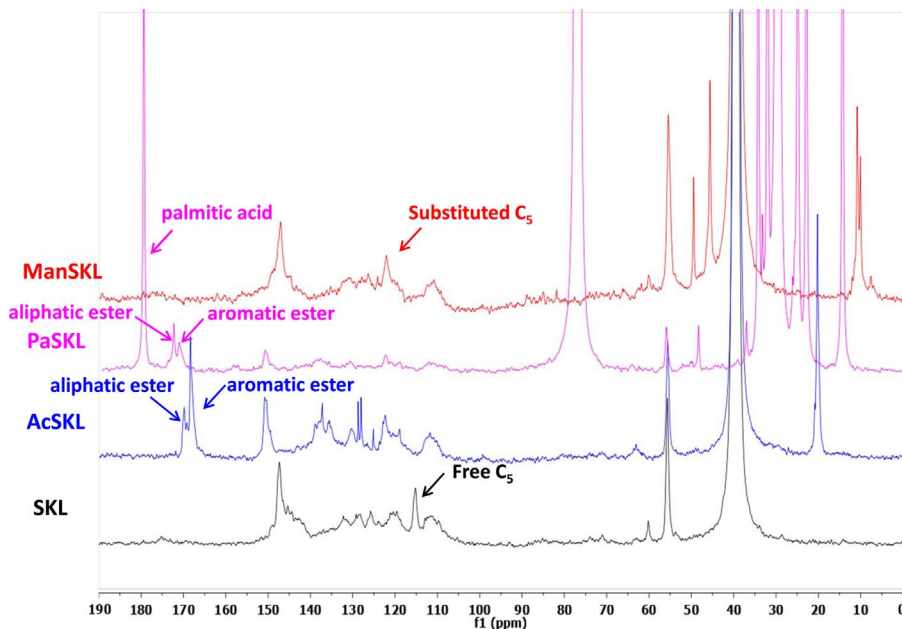
Fig. 3. FTIR spectra of original SKL, AcSKL, PaSKL and ManSKL.

content increased obviously to 2.86 mmol/g from the starting content of 0.46 mmol/g. Similar to acetylation, the original carboxylic groups should remain intact after palmitoylation, and the increased amount of carboxylic groups should come from the free palmitic acid generated from the decomposition of unconsumed palmitic anhydride. In contrast to the complete removal of all acetic acid by evaporation after the above-mentioned acetylation, there is no simple method that can be used to purify the PaSKL product from free long-chain palmitic acid. In other words, the final prepared lignin derivative (PaSKL) is in fact a mixture of palmitoylated lignin and free palmitic acid. A similar mixture was earlier reported by Tamminen et al. using tall oil fatty acid for esterification (Tamminen et al., 2013). As shown in the FTIR spectra (Fig. 3), compared to the starting SKL, the intensity of the signals at approximately 3300 cm^{-1} , generated from various OHs, again vanished almost completely. Two carboxyl peaks also emerged at 1765 and 1734 cm^{-1} , generated from the esterified form of aromatic and aliphatic alkanoxo groups, respectively (Glasser and Jain, 1993). The structural changes were verified by ^{13}C NMR analysis, in which the carboxyl groups originating either from esters from both aromatic and aliphatic alkanoxo groups or from free palmitic acid were observed by

signals at approximately $170\text{--}180\text{ ppm}$ (Fig. 4). Again, esterification occurred at both the C_α and C_γ positions and C_4 position on the aromatic rings. Similarly, corresponding signal shifts were observed in the 2D NMR spectra for C–H at the C_α (disappeared), C_γ and C_5 positions (Fig. 5). Fortunately, the presence of free palmitic acid in the PaSKL product is acceptable, as it is also very hydrophobic and beneficial for improving the coating effect. Therefore, PaSKL was used without further purification.

The second type of functionalization conducted is the Mannich reaction using formaldehyde and diethylamine which primarily occurs at the phenolic G units and a dialkylaminomethyl group is introduced at the C_5 position of each reactive (phenolic) aromatic ring by formaldehyde together with an amine (here with diethylamine) (Fig. 2). The nitrogen atom presents as an organically bound dialkylaminomethyl group, which will be a slow release form of N (Chien et al., 2009).

As shown by ^{31}P NMR analysis (Fig. 1), completion of the Mannich reaction was shown by the complete disappearance of all phenolic G units ($140.2\text{--}139.0\text{ ppm}$) and an emergence of phenolic condensed units after attack of the C_5 site by the dialkylaminomethyl group

Fig. 4. ^{13}C NMR spectra of original SKL, AcSKL, PaSKL and ManSKL.

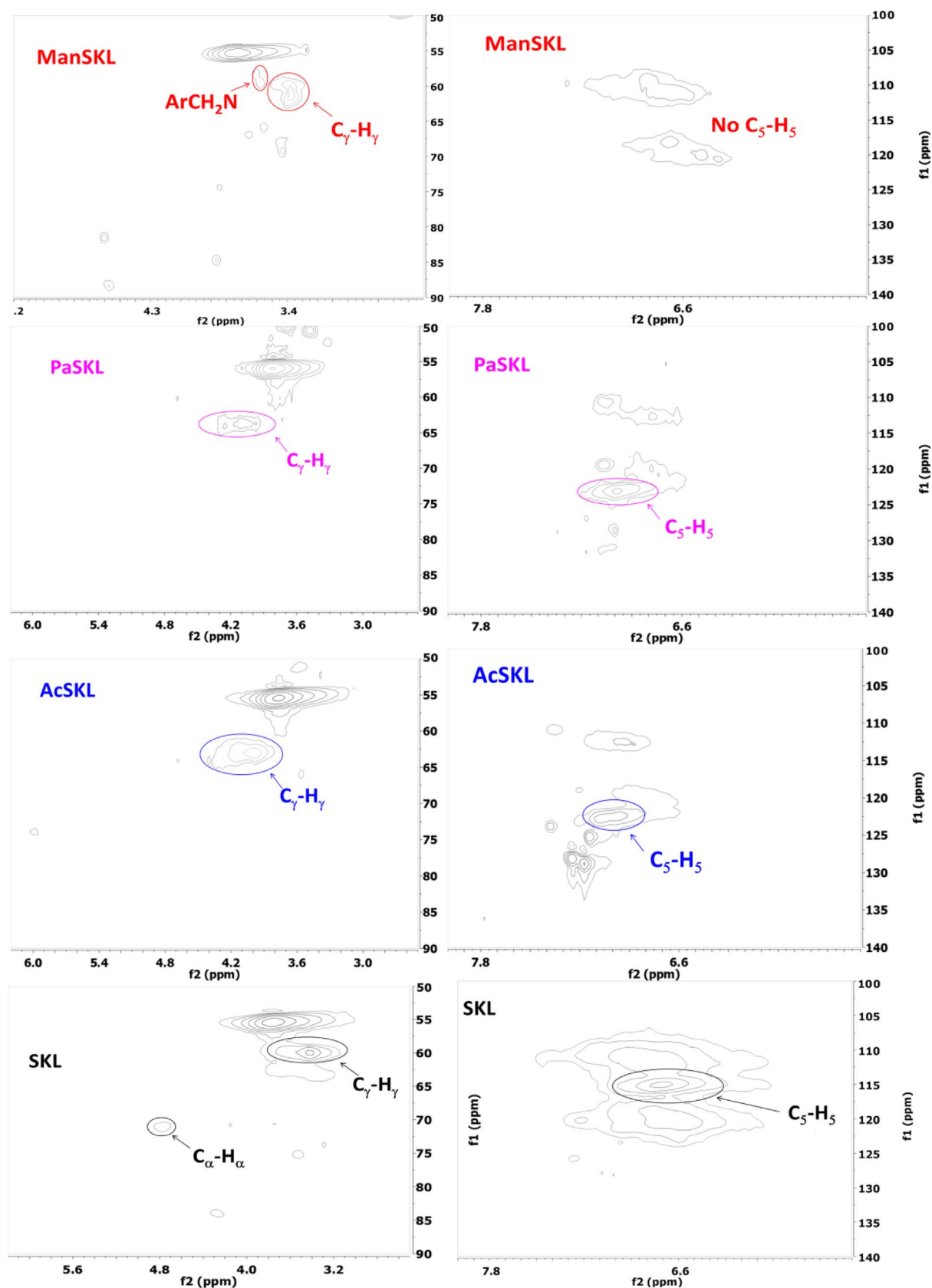


Fig. 5. Enlarged 2D NMR spectra at side chain (left column) and aromatic (right column) C–H signal regions of original SKL, AcSKL, PaSKL and ManSKL.

(145.4–143 ppm). Due to the reaction specificity (Fig. 2), all other hydroxyl groups remained generally intact – the peaks of the aliphatic and carboxylic OHs remained intact (Fig. 1) – and only certain

reductions in the content of aliphatic hydroxyls and carboxyls and the total amount of the various phenolic hydroxyls were observed (Table 1). Preservation of the multi-phenolic groups will ensure the

preservation of the lignin inhibitory effects against urease hydrolysis and microorganism nitrification, as these are the groups involved in these effects (Lodhi and Killingbeck, 1980). Due to the introduction of the diethylaminomethyl group, a new $-\text{N}(\text{CH}_2\text{CH}_3)_2$ signal was observed in the FTIR spectrum at $\sim 3000\text{ cm}^{-1}$ (Fig. 3). The ^{13}C NMR spectrum confirmed complete substitution at almost all C_5 positions, such that the original C_5 carbon signals shifted downfield to a higher ppm value (Fig. 4). 2D NMR (Fig. 5) verified that no C_5H_5 signal remained, but instead a merged ArCH_2N structure existed (see (Du et al., 2014)). By direct determination of the total nitrogen, the N content in ManSKL was 2.5 wt%, similar to the value reported previously (Du et al., 2014).

3.3. Coating urea

3.3.1. Coating technique

In this study, PLA was used as the basic material for coating. In the literature, PLA has been used in a chloroform solution for dip-coating on metal material (Shi et al., 2015). Here, we used a similar but less toxic solvent, dichloromethane (DCM). Although dissolution in DCM was very slow, a clear solution of 60% (w/v) concentration was successfully prepared after stirring overnight at room temperature. Since lignin and lignin derivatives are insoluble in DCM, to conduct PLA and lignin complex coating, it was dissolved in dioxane and then mixed with the PLA/DCM solution. Dioxane was found to be well miscible with DCM, such that the mixed DCM-dioxane solution was perfect for conducting dip-coating.

3.3.2. Preparation and evaluation of cast films

In the dip-coating process, the coating layer is formed in a similar manner to film formation from solution. Therefore, the PLA-lignin complex coating was first evaluated by film casting for coating capacity and coating quality. The film formed from a 60% DCM solution of the basic PLA material is a uniform film, as seen by optical observations (Fig. 6a). However, under SEM analysis, many pores are visible on the nanometer scale (Fig. 6b). Further image analysis revealed that PLA cast film has pores with an average pore size of $410 \pm 155\text{ nm}$. These pinholes will lead to water or steam penetration and then urea dissolution and transport. Apparently, PLA alone could not form a good film. From the PLA and lignin complex in DCM-dioxane solution, unmodified SKL was found to phase-separate from PLA in the cast PLA-lignin film (Fig. 6c), forming pores with even larger sizes ($1133 \pm 553\text{ nm}$) than those in the PLA film (Fig. 6d). However, after functionalization by esterification, both esterified lignins formed optically homogeneous films with PLA (Fig. 6e and 6g). Under SEM observation, however, AcSKL caused the formation of large-sized chunks (Fig. 6f) with pores in size of $775 \pm 348\text{ nm}$. In contrast, a very homogeneous layer without any visible pores was formed from PaSKL (Fig. 6h). Therefore, palmitoylation functionalization resulted in an excellent reinforcement effect in PLA film formation. On the other hand, after functionalization by the Mannich reaction, a similar perfect reinforced film was formed from the PLA-ManSKL complex; a very homogeneous (Fig. 6i) and pore-free layer was formed (Fig. 6j). The starting SKL has a molecular mass distribution of $M_w = 5500\text{ g/mol}$. After functionalization, both PaSKL and ManSKL still have molecular masses in a similar range. Compared to the PLA, whose M_w is 140567 g/mol according to the manufacturer, these lignin derivatives apparently functioned as a plasticizer for PLA, which improved the film formation and quality.

The desired function of the coating layer is as a physical barrier to block water. This capability is closely related to the film hydrophobicity, which can be evaluated by contact angle values (Table 2). It could be seen that the original PLA film is somewhat hydrophilic (47.50°). Since lignin is essentially hydrophobic due to the presence of multiple aromatic rings in its structure, the PLA-SKL film becomes obviously more hydrophobic (79.39°). After functionalization, good

hydrophobicity is achieved by esterification. Due to the presence of longer aliphatic chains in palmitic acid compared with those in acetic acid, a higher hydrophobicity is observed for the PLA-PaSKL film (87.01°) than the PLA-AcSKL film (70.64°). The lower contact angle for PLA-AcSKL film than PLA-SKL film should be due to the formation of large-sized chunks in the former one, which is confirmed by SEM observation as shown in Fig. 6f. After the Mannich reaction, the PLA-ManSKL film is more hydrophobic (65.52°) than the PLA film (47.50°) but less hydrophobic than the PLA-SKL film (79.39°) since the introduced diethylaminomethyl group (Fig. 2) is somewhat hydrophilic.

3.3.3. Dip-coating results

Since lignin aggregation (as chunks) was observed in the PLA-AcSKL film, which has also lower contact angle than the PLA-PaSKL film, our dip-coating experiments focused on the PLA-PaSKL and PLA-ManSKL coatings, which represent different types of functionalizations, using a PLA coating for comparison. Generally, the simple dip-coating performance resulted in good attachment of the coating layer on the surface of urea pellets, as shown by the cross-sectional image of a coated urea pellet (Fig. 7a). A typical coated urea pellet is shown in Fig. 7b, and it could be seen that the coating layer is very compact and similar to the cast films. The wall layer is rather homogeneous and well distributed. The coating process was repeated several times, and the quantity of the coating was found to be dependent on the number of coating cycles. This tendency is, in general, irrespective of the type of lignin derivative used (Table 3).

In fact, the coating layer was so compact and strong that the layer survived after immersion in water and complete release of the urea core (Fig. 7c) (see section 3.4). After complete urea release, which removed the entire urea core, SEM images of the surviving coating layers clearly revealed that the wall thickness is dependent on the number of coating cycles, and no layer separation was found between different coating layers, irrespective of the coating mixture components (examples of the PLA coating are given in Fig. 7d and 7e and the PLA-PaSKL coating are given in Fig. 7f and 7g). In other words, the coating is one single layer, and its quantity or coated wall thickness could be tuned by adjusting the number of dip-coating cycles. To our best knowledge, this is the first time that a single-layer coating using a PLA-lignin blend has been reported.

Upon calculation, the urea core itself contains 46.6% N. In the PLA-lignin complex, 9.1% of the coating mixture was from lignin. Therefore, for the coated urea with the thickest shells, i.e., 51–81% coating (Table 3), the added lignin quantity is 5–8% of the urea weight. When ManSKL is used, since ManSKL contains 2.5% N, the coating shell will contain 0.23% organic N. For the coated urea with the thickest shells, therefore, approximately 0.25–0.40% of the total N will be organically bound N.

3.4. Nitrogen release

In this study, immersion testing using water, one of the simplest and most commonly used methods in the literature (Chen et al., 2002; Chen et al., 2009; Wang et al., 2014a) and in several national and international standards (see (Carson and Ozores-Hampton, 2012)), was applied to evaluate the release performance of the PLA-lignin complex coated fertilizers. It can be seen from Fig. 8 that the complete release of bare urea took only 50 s (seen also by the disappearance of the urea pellets). The PLA coating delayed the complete release to 150 s, i.e., three times as long as that without the coating. Esterified PaSKL delayed the release to approximately 1500 s, i.e., 10 and 30 times as long as that of the PLA coating and bare urea, respectively. The Mannich reaction product ManSKL also slowed the urea release more than the PLA coating reference, taking approximately 1000 s, or 6.7 times as long. This release is 20 times as long as that of bare urea. Here, when PLA was used alone as the coating material, the urea release delay is due to the physical barrier property of PLA coating. The delay differences from the PLA-

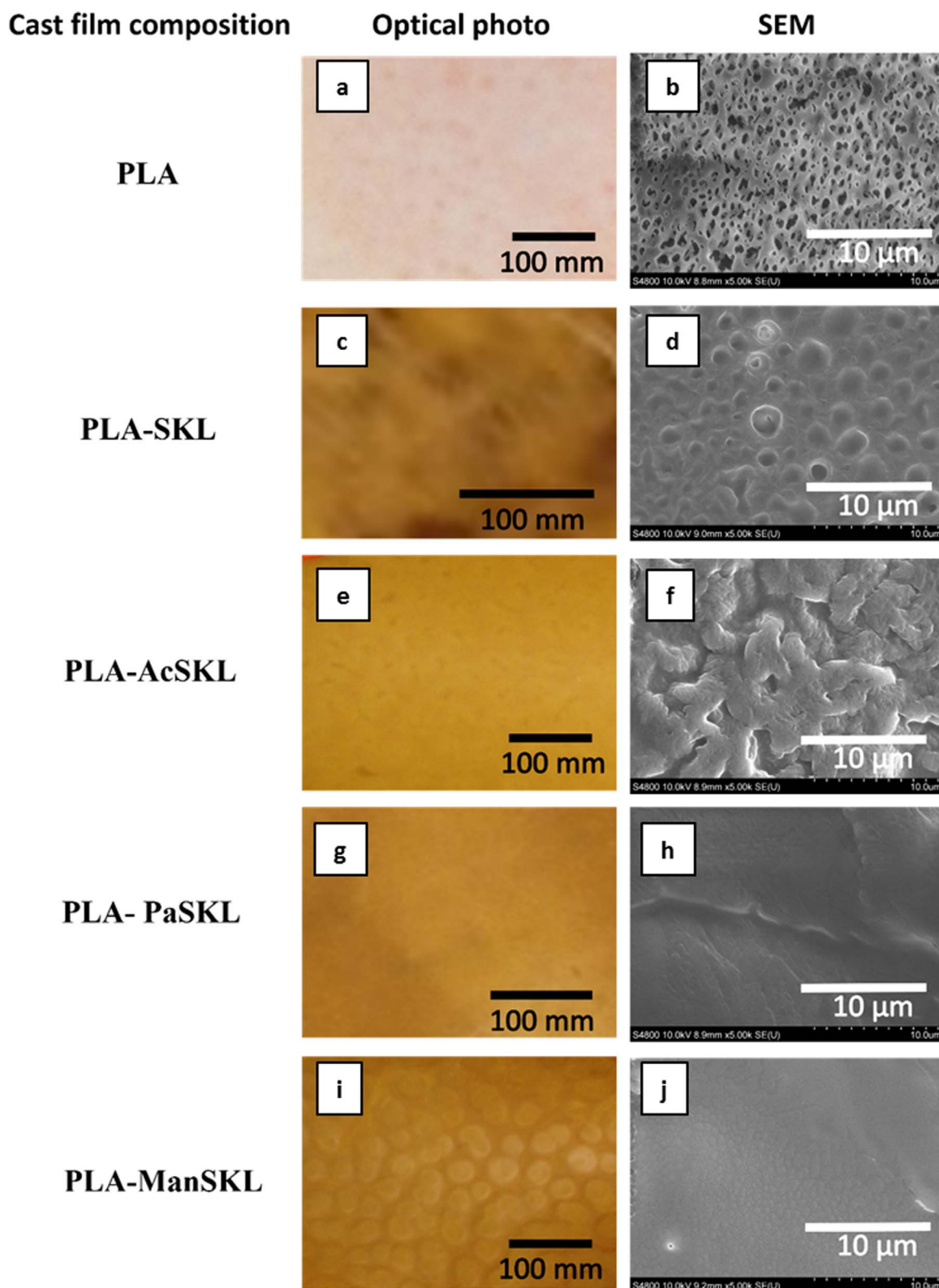


Fig. 6. Optical microscope (left column) and SEM images (right column) of cast films from PLA, PLA-SKL, PLA-AcSKL, PLA-PaSKL and PLA-ManSKL.

PaSKL and PLA-ManSKL complex coatings are at least partly originating from the differences in actual thickness of the coating, imperfections in coating surface and chemistry of the coating material. In terms of different chemical functionalizations, after the palmitoylation, long aliphatic chain was introduced to the lignin structures and the largely increased hydrophobicity is responsible for the longest release delay

(1500 s) observed. On the contrary, although the Mannich reaction caused the lignin hydrophobicity deterioration due to the introduced hydrophilic diethylaminomethyl groups, the aminated lignin, ManSKL, prepared was thought to have an advantage as itself being a slow-release N since it contains amine groups with still high molecular mass sizes (Ariyanti et al., 2013; Du et al., 2014).

Table 2
Contact angles (°) of the films cast from PLA and the PLA-lignin complex^a.

	Neat PLA	PLA-lignin complex			
		Original SKL	After esterification		After Mannich reaction
		PLA-SKL	PLA-AcSKL	PLA-PaSKL	PLA-ManSKL
Mean	47.5	79.39	70.64	87.01	65.52
SD ^b	0.99	3.5	1.04	1.05	5.09

^a all contact angles were read after 60 s when the readings were stable.

^b standard deviation.

The lignin functionalization conducted in this study could be expected to also influence the later steps in soil. One important effect from the lignin structure is the inhibition of urease hydrolysis and nitrification by soil microorganisms. In the literature, a lignin coating was found to be capable of providing a 12% decrease in the release rate of bare urea in soil and give a higher urea efficiency in corn cultivation due to lignin's inhibition of nitrification (Peng et al., 2004). The mechanism of lignin's inhibition of nitrification includes hydrogen-bond formation between lignin hydroxyl groups and ammonium, which is related to the polyphenol structures in lignin (Sabahi et al., 2014). Lignin could also reduce N₂O emissions by approximately 83–96%, thus having potential as a fertilizer amendment to improve the N fertilizer efficiency (Huang et al., 2003). In addition, the utilization of lignin for coating urea can lead to many other advantages. For example, the

Table 3
Amounts of coating wall (percentage over urea core weight) from different PLA-lignin derivative complex coatings after repeated cycles of dip-coating.

Coating repetition times	Coating quantity (%)
1	6–24
2	26–51
3	42–63
5	51–81

hydrophobicity of lignin observed in this study would help prevent PLA from undergoing hydrolysis, as hydrolytic instability is a critical PLA weakness during application (Södergård and Inkinen, 2011). As reviewed by Gul et al. (2014), lignin structures can influence the soil microbial community structure to facilitate the control of soil quality through the reduction of greenhouse gas emissions from soil, the retention of soluble nutrients, the promotion of soil aggregate formation and stabilization, and the bioremediation and detoxification of natural and man-made organic pollutants.

Generally, several improvements could be made in the future to further slow the release of the coated urea reported in this study. As was found, crack channels (Fig. 7c) are mainly responsible for urea release. More research should be conducted to eliminate these cracks or prevent their formation. This could be tested by controlling the speed of solvent evaporation during the dip-coating procedure or conducting annealing to prevent the formation of cracks and pores (Devassine et al., 2002). Sealing the coating layer by the inclusion of wax in the coating complex

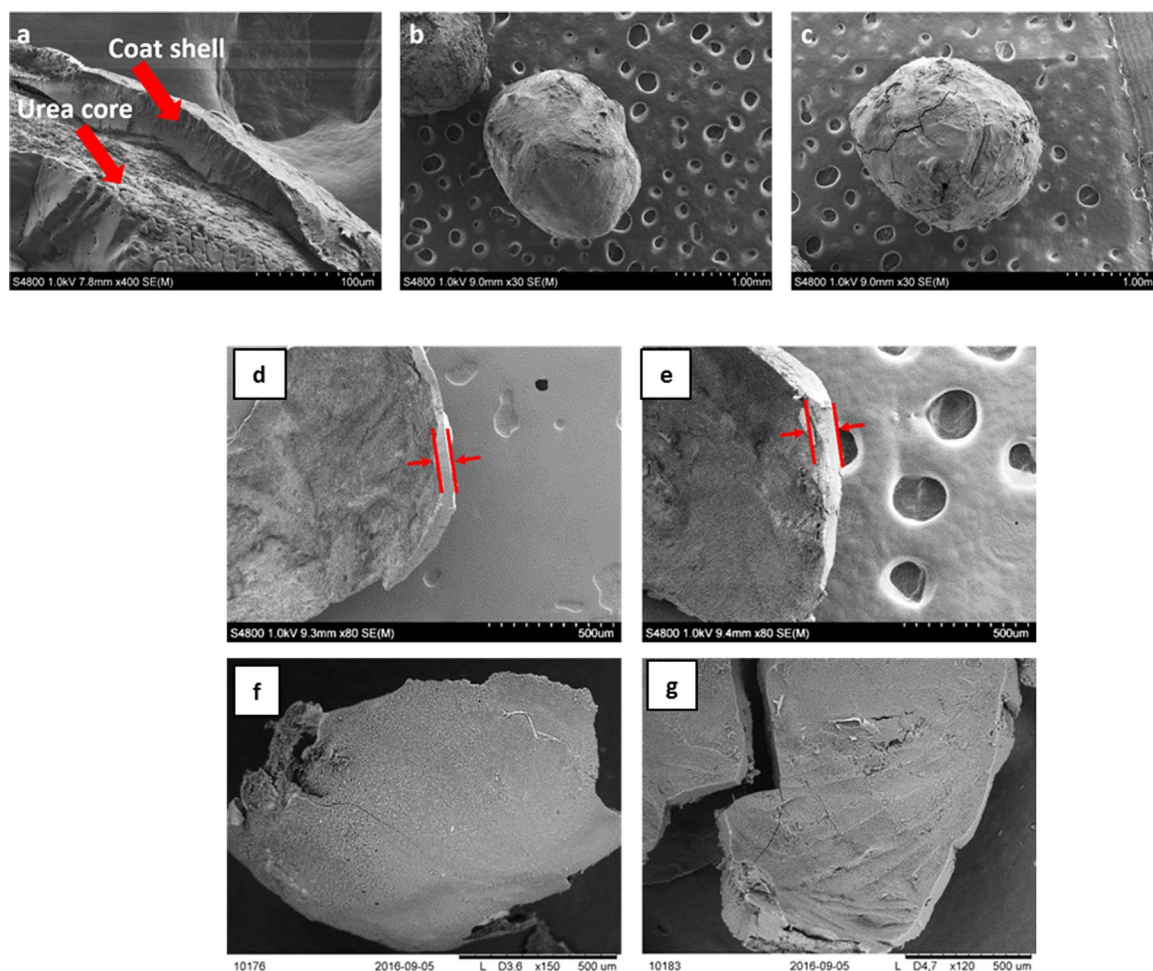


Fig. 7. Typical examples of scanning electron micrograph (SEM) images of a PLA-lignin complex-coated urea pellet, showing a coating layer over urea core (a) and the coated pellet before (b) and after (c) urea release to surrounding water as well as of the residual coating layers after complete urea release. 3 runs (d) or 5 runs of PLA coating (e), and 1 run (f) or 3 runs of PLA-PaSKL coating (g) with a comparable thickness with each other.

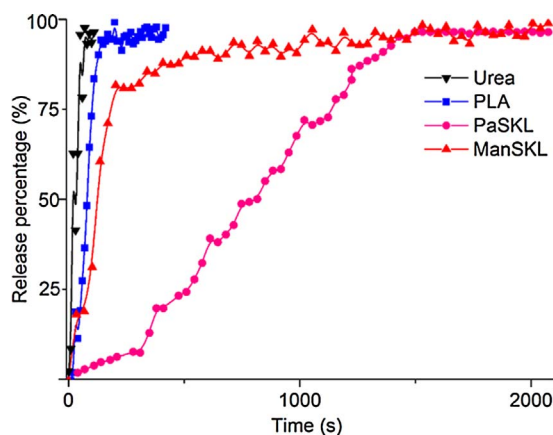


Fig. 8. Results of the circulating water release test for urea release percentage from coated urea pellets after determination using UV at 200 nm. The release percentage was relative to the highest possible reading.

composition could also be tested (Cui et al., 2013; Hudson and Woodward, 1995; Kakoulides and Valkanas, 1994). In addition, the content of organically bound N compared to the total N content in the coated fertilizer is still low. An increase in the ManSKL coating amount could be considered.

4. Conclusions

Industrial lignin was functionalized by esterification or the Mannich reaction to a ~100% completion. Using dip-coating, PLA-functionalized lignin complex-coated urea can be simply prepared with adjustable coating thickness and high coating efficiency. The use of functionalized lignin, especially palmitoylated lignin, obviously delays the release of the urea core in water, and utilization of the Mannich reaction-functionalized lignin supplies additional organically bound nitrogen to the fertilizer. Therefore, the coating conducted in this study not only constructs a physical barrier to delay urea dissolution but also supplies chemically slow-release, organically bound nitrogen, and biological stabilization effects, such that the coated urea is a simple but highly efficient fertilizer.

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